LUCA PIERGENTILI – MEng Aerospace Engineering Student

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ENGINEERING PROFESSIONAL EXPERIENCE

CAMO Engineering Intern – Avia Seven & Marathon Airlines, Athens, Greece **June – September 2026**

Continuing-airworthiness organisation and airline operating Embraer E-Jets and Legacy 600 business jets.

– Assessed Embraer E190 and Legacy 600 **Service Bulletins** for fleet applicability and safety impact. Identified 247 applicable E190 SBs, presented my findings and **planned their implementation** with the CAMO manager.

– Acquainted myself with the **Aircraft Maintenance Programme (AMP)** of one of Avia Seven's E190s and performed a cross-check against Embraer's Maintenance Planning Document (MPD). Logged **16 discrepancies** and informed the CAMO manager who issued corrections.

– Built an **Excel tool** to merge two AMPs of an aircraft transferring from another CAMO to Avia Seven, **automatically flagging duplicate and missing tasks** and checking intervals and thresholds against Embraer's MPD and logged flight hours.

– **Categorised Trieste airport** (Category B) for the airline's E-Jet and Legacy 600 operations, assessing terrain, approaches and weather.

Product Design Engineer (Industrial Placement) – Andel Ltd, Barnsley, UK **March – August 2025**

Leak-detection and flood-protection products for industry and utilities.

– **Self-initiated the redesign** of Andel's flagship: BundGuard, replacing welded brackets with modular bolted attachments to support all variants using one sensor body. Reduced sensor footprint by 67%; prototyped and secured **senior management approval** for adoption.

– Redesigned the FlowStop 2 valve enclosure: made it **20% more compact**, with a larger flat back for adhesive mounting; valve now only fits the right way round and no longer rattles. Now **printed in-house on SLS** instead of outsourced.

– Redesigned the Gully Flood Sensor enclosure for **fast installation without road closures**; field-tested and **approved for commercial use**.

– Built **parametric CAD models** for three product families (BundGuard, Point Sensor, CellarGuard) to aid the 3-person R&D team to rapidly develop product variants. Also designed **manufacturing jigs** and build documents for the Production team to simplify manufacturing.

Design Engineer (Industrial Placement) – ChargePoint Technology, Liverpool, UK **August 2024 – February 2025**

Containment and sterile-transfer valves for pharmaceutical manufacturing.

– Automated the company-wide product and item databases by decoding product codes with Excel functions to provide technical product information, this led to company-wide facilitated access to information. Delivered the first version in **1 month against a 6-month plan**.

– Investigated a valve failure that broke sterile containment: used creative thinking to design **pressure tests** that recreate the failure conditions, **identified the failure mechanism** and proposed design fixes to the R&D team.

– Followed customer design brief and **halved the size** of a product's sensor box, allowing full product functionality in space-limited plants.

– Designed a stainless-steel test frame for a 100 kg product (performed **beam theory calculations and FEA**) and a lifting-eye handle later **used on a customer order**. Learnt how to design structures based on off-the-shelf extrusions and design-for-welding principles.

EDUCATION

UNIVERSITY OF BATH **September 2022 – June: 2027**

MEng (Hons) Aerospace Engineering (5 Year Master's Course with Year in Industry) – ***Predicted grade: 1st Class Honours***

Key Modules – Aerodynamics, Aerospace Sciences, Aerospace Structures, Aircraft Performance and Design, Aircraft Propulsion, CFD, Rotorcraft Engineering, Spacecraft Engineering, Stability & Control, Supply Chain and Management.

ECOLINT – INTERNATIONAL SCHOOL OF GENEVA – La Grande Boissière Campus **September 2009 – June 2022**

International Baccalaureate – ***Graduated class of 2022 with Distinction and Merit***

Key Subjects – Higher Level Mathematics (7/7), Higher Level Physics (7/7), Standard Level Economics (6/7) and Extended Essay (A)

SPOKEN LANGUAGES: Italian – C2, English – C2, French – B2, German – A1

ENGINEERING PROJECTS

Final Year Project – Analysis of Serrated Wingtip Devices **October – Expected completion: May 2027**

– Working on a **self-initiated individual project** investigating the aerodynamic effect of sine-wave serrations on the top and trailing edges of wingtip devices. The aim is to weaken wingtip vortices and **reduce drag without losing lift**, for wide-body aircraft such as the A350.

– **Ansys CFD** will be used to run simulations for each wingtip configuration and **wind tunnel testing** will validate the simulation results.

LEGO Model Making **June 2019 – Present**

– Designed a 1,000+ piece Rolls-Royce Trent XWB replica, using LEGO CAD Software: Studio. Model recreates the fan, compressor and turbine stages, internal engine structure, openable cowling and **working thrust reverser**. Gained insight on the engine's inner workings.

– Designed Star Trek starship models from reference images, matching proportions to scale and ensuring structurally sound models within the **design constraints of LEGO**, improving my design problem-solving. Ordered the parts and built some of them as display pieces.

– Found a flaw in a finished build that led it to split in half, redesigned the section to **increase structural integrity** while improving source material faithfulness and built the refined version; later applied the same judgement to light, low-material 3D-printed parts at Andel.

Group Business and Design Project – Project Manager & Technical Integrator · 76% (First) **February – May 2026**

– **Led an eight-person team** designing a commercially competitive autonomous fixed wing VTOL-capable agricultural spraying UAV. Eased the flow of ideas during initial concept design and provided clear and focused objectives and schedules during design development.

– Was responsible for **design integration** and owned aircraft configuration design, component placement, COG location, requirements tracking and parametric CAD model. Introduced an end-of-project **day-by-day plan**, guiding design freezes, leading to early submissions.

Wing Design – MATLAB Design Optimisation · 72% (First) **October – December 2025**

– Wrote a MATLAB script to **mass-optimise a CFRP wing panel** for a Saab 340, sizing the laminate against design loads and reserve factors and automatically selecting the optimal layup pattern according to industry laminate rules. Gained insight in composite design.

Team Bath Drones – Design and Make Activity **October – December 2025**

– **Led a five-person team** designing a fixed-wing drone to drop 4 kg of water, performed **aerodynamic analysis** and lift and drag estimates.

Machine Design – Team Lead · 71% (First) **February – May 2024**

– **Led a four-person team** designing an autonomous control-rod module assembly machine, guided concept ideation and coordinated the project. **Performed FEA** on critical parts, selected and integrated off-the-shelf components and created a full machine CAD model.

Tailfin – @home Design Challenge **June 2023**

– Designed a **tool-less, length-adjustable** AeroPack seat-post connector to needs gathered from cyclists; it withstood bending stress.

Other Design Projects **February 2019 – May 2025**

Extended Essay – Designed and built a guitar test rig that **disproved the Tone-Wood theory**. Received top marks for the research essay.
Guitar Modification – Modified an electric guitar by **redesigning the electronics**, allowing 10 sound combinations, replacing the hardware and the neck, and created 20 design concepts for the guitar's paint job. Created a sealed paint-booth to spray-paint the guitar.

TECHNICAL SKILLS

CAD Software Proficiency: Advanced user of **Inventor Professional** (Bath), **Fusion 360** (Andel) and **SolidWorks** (ChargePoint) – advanced geometry modelling, parametric design, motion-enabled assemblies, design accelerators and BS 8888 technical drawings.

Analysis and Simulation: Good proficiency with **MATLAB** and **Python** – numerical methods, design optimisation, flow field solver and GUI design. Intermediate user of **Ansys CFD** – laminar and turbulent flow simulation. Early user of **XFLR5** – basic aerodynamic analysis.

Engineering: Systems integration, technical coordination, problem solving, creative thinking, product design and project management

Office Tools: Advanced user of: Word, PowerPoint, Visio, Excel (advanced functions and automation) and their Google equivalents.

INTERESTS and ACTIVITIES

Competitive Sports

Played volleyball in university teams and for Liverpool VC, demonstrating teamwork and team spirit. Also assisted in training planning and drills at Liverpool VC. Was the starting goalkeeper and team captain of Veyrier's under 18 football team, exercising leadership at a young age. Acquired a sailing licence and worked as an instructor, developing communication skills to better teach sailing to a diverse audience.

Music and Performing Arts

Independently learnt how to play 4 instruments: piano, guitar, bass, and drums. Recorded, produced and published original music on YouTube, while autonomously gaining experience with Logic Pro. Took responsibility for the school's recording studio, and took part, as leading role, in 2 school productions and performed with two singing ensembles, recording and producing covers with them.

REFERENCES

Prof Alborz Shokrani – Department of Mechanical Engineering

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